01 | How spintronics works

Using electrons’ charge and spin

Spin is related to an electron’s magnetic properties. MRAM, a representative example, is memory that retains stored information when power is turned off.

Memory element: MTJ (magnetic tunnel junction)

Magnetic layer | ultrathin insulating film | magnetic layer. Uses the “tunneling effect,” in which electrons pass through the insulating film. Resistance changes according to the magnetization directions of the two magnetic layers.

From writing to reading

① Current or other means changes one layer’s magnetization to write 0 and 1.
② Parallel magnetization gives low resistance; antiparallel gives high resistance.
③ This resistance difference is used to read 0 and 1.
No current is needed to retain the record, helping reduce power use.

02 | Japanese research institutions and companies

Connecting basic research, manufacturing equipment and products

Basic research | AIST and Tohoku University

AIST developed MTJs using magnesium oxide (MgO). Tohoku University also advances research on magnetic materials and spin control. Japanese research contributes to MRAM’s foundational technology.

Manufacturing equipment | Canon Anelva and Tokyo Electron

Magnetic materials and insulating films are stacked in extremely thin, uniform layers. Canon Anelva offers deposition and processing equipment for MRAM. Tokyo Electron offers deposition equipment for STT-MRAM.

Products | Renesas Electronics and TDK

Renesas commercializes microcontrollers with embedded MRAM. TDK offers sensors using the TMR (tunnel magnetoresistance) effect. MRAM stores information; magnetic sensors detect magnetic fields.

03 | Commercialization: progress and challenges

Fields reaching products, and fields still in R&D

Areas moving into practical use

Microcontrollers with embedded MRAM and TMR magnetic sensors are already commercial products. MRAM retains data after power is off. Some Renesas microcontrollers adopt TSMC’s MRAM technology.

Technology and manufacturing challenges

Challenges include larger capacity, manufacturing cost and lower write power. Speed, retention and endurance must be balanced for each use. Thin-film uniformity and mass-production yield are also important.

Next-generation technology and wider applications

Faster next-generation MRAM and computing applications are under R&D. Performance and cost determine how it is used alongside existing memory. It has not been decided that it will replace all DRAM.